

Financial Document Management System

1. Introduction

This project is developed using FastAPI and Python. The main purpose of this project is to upload and manage financial PDF documents. The system also performs semantic search using RAG technology. Users can search important information from uploaded documents easily.

2. Architecture

The project follows a simple backend architecture. FastAPI is used for API development. FAISS is used as a vector database. Sentence Transformers are used to generate embeddings for semantic search.

3. APIs

The project contains APIs for user registration, login, document upload, document retrieval, document deletion, and semantic search. All APIs are tested using Swagger UI.

4. Database Design

The project stores document metadata such as document_id, title, company_name, document_type, uploaded_by, and created_at. FAISS vector database is used for storing embeddings.

5. RAG Workflow

First, the user uploads a PDF document. The system extracts text from the PDF and divides it into smaller chunks. Embeddings are generated from the text and stored in FAISS. When the user searches a query, the system returns the most relevant information.

6. Features

The project includes user authentication, PDF upload, semantic search, document management, and Swagger API documentation.

7. Advantages

The system provides fast document retrieval and better search accuracy using AI-based semantic search. It reduces manual effort and makes document searching easier.

8. Conclusion

The Financial Document Management System successfully performs PDF processing and semantic search using FastAPI and RAG technology. The project is useful for managing financial documents in a simple and efficient way.